MILITARY SPECIFICATION

INSTRUMENTS, AIRCRAFT, GENERAL
SPECIFICATION FOR

This specification has been approved
by the Naval Air Systems Command,
Department of the Navy.

1. SCOPE

1.1 Scope - This specification covers the general requirements for the design and manufacture of aircraft instruments for operation primarily in piloted aircraft. The detail performance and test requirements for a particular aircraft instrument shall be as specified in the detail specification for that instrument.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

BB-N-411	Nitrogen, Technical
DD-G-451	Glass, Flat and Corrugated, for Glazing, Mirrors, and Other Uses

Military

MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-I-6181	Interference Control Requirements, Aircraft Equipment

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:34Z
Check the source to verify that this is the current version before use.

Military (continued)

MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-S-20708	Synchros, 60 and 400 cycle, General Specification for
MIL-R-23417	Resolvers, Electrical, General Specification for
MIL-L-25467	Lighting, Integral, Aircraft Instrument, General Specification for
MIL-C-26482	Connectors, Electric, Circular, Miniature, Quick Disconnect
MIL-B-27497	Bearing, Jewel, Sapphire or Ruby, Synthetic

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-280	Definition of Terms for Equipment Divisions
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MS33558	Numerals and Letters, Aircraft Instrument Dial, Standard Form of
MS33585	Pointer, Dial, Standard Design of Aircraft Equipment
MS33737	Nuts, Self-Locking, Clip-In Type, Instrument Mounting

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:34Z
Check the source to verify that this is the current version before use.

PUBLICATIONS

Air Force - Navy Aeronautical Bulletins

ANA - 400	Electronic Equipment: Aircraft and Guided Missiles, Applicable Documents
ANA - 418	Differential Pressures

Naval Air Systems Command, Requirement

AR-5	Microelectronic Devices Used in Avionics Equipment, Procedures for Selection and Approval of

2.2 Availability of documents - When requesting any of the applicable documents, refer to both title and number. Copies of this specification and applicable documents may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code 1051, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

3. REQUIREMENTS

3.1 Materials, parts and processes - Materials, parts and processes shall conform to the requirements of MIL-E-5400, the specifications listed in ANA Bulletin No. 400 and the requirements listed herein. Electron tubes shall not be used. Where there are no suitable standard parts or materials such parts and materials shall be used which are of the best commercial quality, of the lightest practicable weight, entirely suited for the purpose and readily available. However, only nonflammable materials shall be used. Commercial parts such as screws, bolts, nuts, cotter pins, etc., may be used provided:

(a) They can be replaced by a standard (MS or AN) part without alteration.

(b) The corresponding standard part numbers are referenced in the parts list and on the contractor's drawing.

3.1.1 Part testing - No part shall be used in the equipment without evidence of the part having been tested. The test results may be available from a data exchange program, from previous contracts, or if not so available, the parts must be tested under this contract requirement. Unless required by the contract or equipment specification, the submission of test information on parts is not required, but the contractor shall be able to produce test results when requested by the procuring authority.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:34Z
Check the source to verify that this is the current version before use.

3.1.2 Microelectronic devices - When used, microelectronic devices shall be selected and approved in accordance with AR-5.

3.1.3 Metals - Metals shall be of the corrosion resistant type or shall be suitably protected as specified in MIL-STD-454 to resist corrosion due to fuels, salt spray, or atmospheric conditions to which the instrument may be subjected when in storage or normal service life.

3.1.4 Magnetic materials - Magnetic materials shall not be used except where magnetic materials are essential.

3.1.5 Jewel bearings - shall be in accordance with MIL-B-27497.

3.1.6 Mounting Flange and Bezel - When a MS-28055 attachable mounting flange is furnished with an indicator, it shall remain a fixed part of the case when installed. If the case is a MS-33638 and has a removable bezel, the bezel shall be held in place by means of screws secured by lock washers to make it a fixed part of the case. Removal of the bezel shall permit replacement of the cover glass.

3.1.7 Connectors - The electrical connectors used shall be bayonet coupling type in accordance with MIL-C-26482, and MIL-STD-454, Requirement 10, and of the type specified in the detail specification.

3.1.8 Synchros - shall be selected in accordance with MIL-S-20708.

3.1.9 Resolvers - shall be selected in accordance with MIL-R-23417.

3.1.10 Electrical Indicating Instruments - Electrical indicating instruments shall be of a sealed type and shall be connected in circuits so as not to constitute a personnel hazard in the event of failure.

3.2 Design and construction -

3.2.1 General - The detailed mechanical and electrical design of the instrument shall be accomplished by the contractor in accordance with MIL-E-5400, subject to the requirements of this specification and any specification to which it is subsidiary. The requirements of this specification are detailed only to the extent considered necessary to obtain the desired mechanical and electrical characteristics, performance, and permanence of the same. The design layout and assembly of the units and their component parts shall be such as to facilitate quantity production and to result in minimum size and weight. Electron tubes shall not be used.

3.2.2 Fabrication - Boxes, cases, shields, and compartment walls shall be made by casting, drawing, or bending, and welding, except when ease of servicing of the equipment requires that a removable panel construction be used, or when the applied stresses dictate the use of a strong aluminum alloy which does not provide a good weld; for such parts, riveting or bolting may be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:34Z
Check the source to verify that this is the current version before use.

3.2.3 Maintenance and field testing - The design shall be reasonably simple to facilitate as much as possible disassembly, repair or overhaul service maintenance and reassembly by those tools and items of maintenance equipment which are normally available as commercial standards without requiring special tools and fixtures. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

3.2.4 Cases - shall be made of non-ferrous, low density metal, uniform in texture, having a smooth surface and shall be hermetically sealed or rain and dust tight as specified in the detailed specification. Cases installed in the cockpit area shall be finished with a lusterless black material, Color No. 37038 of FED-STD-595 or black wrinkle finish. Finish of all other cases shall be Color No. 36231 of FED-STD-595. The finishing material shall be of a durable type to withstand usage encountered in service. Indicator cases shall be in accordance with the requirements of the detail specification.

3.2.4.1 Hermetically sealed cases - shall be so constructed that they may be opened, the mechanism removed and replaced and the case resealed at least three times. This shall be possible without the use of any special tool, jig or fixture. Special devices shall be approved by the procuring activity. The sealing of the case shall not be dependent upon any material which will be adversely affected by any atmosphere to which the instrument may be subjected in normal use in military aircraft.

3.2.4.2 Filling medium - for hermetically sealed cases shall be a mixture of 90 percent nitrogen and 10 percent helium, or a mixture of 95 percent helium and 5 percent carbon dioxide, or a mixture of gases approved by the procuring activity. The nitrogen used shall be a quality in accordance with BB-N-411, Grade A, Type I. The filling medium shall be of at least 98 percent purity, free of dust particles and containing not more than 0.006 milligram of water vapor per liter (dew point -65 °C) at the filling pressure. The case shall be filled to an absolute pressure of 1±0.1 atmosphere.

3.2.5 Dials and pointers -

3.2.5.1 Dial plates - Dial plates made of aluminum or aluminum alloys are preferred. The plate shall be securely fastened to the frame of the mechanism in such a manner that it will not loosen or turn when the instrument is vibrated. If screws are used to attach the plate, they shall be so located that they will not interfere with the proper location of any dial marking.

3.2.5.2 Visibility of dial - The pointer, numerals, at least 1/32 inch of the shortest graduations, and all other specified markings on the dial shall be visible from any point within the frustum of a cone whose sides make an angle of 30 degrees with a perpendicular to the dial and whose small diameter is the aperture of the case.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:34Z
Check the source to verify that this is the current version before use.

3.2.5.3 Integral lighting - The integrally lighted instrument shall be in accordance with the requirements of MIL-L-25467 using 5 volts AC or DC.

3.2.5.4 Cover glass - quality shall be in accordance with DD-G-451, Type II, Quality AA. Any flaws permitted by DD-G-451 shall not interfere with reading of the dial. The glass shall not be fluorescent when exposed to ultraviolet light or rays in the 365 millimicrons region.

3.2.5.4.1 Cover glass mounting - The distance between the front surface of the case (mounting flange) and the outside surface of the cover glass and the distance between the dial plate and the inside surface of the cover glass shall be the minimum practicable. For non-sealed cases, the cover glass shall be properly sealed and means shall be provided both for its replacement and to prevent fogging on the inside surface under operational conditions. All visible markings shall be within 0.30 inch distance from the innermost glass surface, preferrably the least practical distance.

3.2.5.5 Dial marking - The scale and other markings on the dial plate shall be as shown in the detail specification including the dimensions of the markings and the colors of the marking materials used. The style and proportions of the letters and numerals placed on the dial plate shall conform to MS33558. The lusterless white markings shall be of Color No. 37875 of FED-STD-595. The lusterless black material shall be of Color No. 37038 of FED-STD-595. The dial markings material shall be of a quality that ensures durability of the markings in extreme service use.

3.2.5.5.1 Zero reference mark - Electrical zero reference mark shall appear on the dial at the angular midpoint of the graduation portion of the dial. The mark shall be located on the edge of the dial scale in order to be visible through the cover glass.

3.2.5.5.2 Numerals - Numerals shall distinctly indicate the graduation to which each applies. Each numeral shall be placed so that the center of mass of the numeral is on the radial line joining the appropriate graduation and the center of the dial. Any confusion resulting in doubt as to which graduation the numeral applies shall be cause for rejection. When several numbers are used in one group, the space between the numerals shall be not less than 1/64th inch. The size of the numerals shall be as specified in the detail specification.

3.2.5.6 Pointers - Pointers shall conform to MS33585. The shaded portion of the pointers shall be finished with lusterless white material of Color No. 37875 of FED-STD-595. The unshaded portion shall be finished with lusterless black material of Color No. 37038 of FED-STD-595. For dual indicator the pointer closest to the cover glass shall be part number MS33585-11. For lighted pointers the pointer shall conform to the above except that they shall be suitable for integral lighting. They shall be as light as practicable and sufficiently rigid to prevent flexing under vibration. They shall be firmly attached to the mechanism but shall be readily adjustable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:34Z
Check the source to verify that this is the current version before use.

3.2.5.6.1 Stops - shall be provided in all instrument to prevent damage to the mechanism.

3.2.5.6.2 Damping - Sufficient damping shall be provided to eliminate excessive oscillation and provide steady indication of the pointer. Instruments employing sensitive type indicators shall be designed to provide practically "dead-beat" operation.

3.2.6 Gears - The use of bevel gears on new designs is not encouraged as experience has shown that considerable service trouble has resulted from their use. The use of simple, direct linkage systems is preferred. When bevel gears are used, they shall be of the "cut" type and so adjusted in assembly with a positive lock adjustment that binding from misalignment or shaft play cannot occur. Tolerances shall be such that binding from temperature extremes or from fuel deposits will be eliminated.

3.2.7 Washers for ceramic and vitreous surfaces - Buffer washers of suitable metallic or fibrous material shall be used between the otherwise facing surfaces of a ceramic or vitric insulator or vitreous enamel resistor and a metal part, whenever practicable. Buffer washers of soft copper or other suitable material shall be used whenever the heat resulting from soldering or like operations during assembly, or internal heat resulting from normal operation of the equipment, is of such degree as to cause deterioration of fiber washers in any manner.

3.2.8 Internal wiring - **Internal wiring shall be color coded, neatly bundled and secured, and accomplished in such a manner that individual wires may be easily traced. All wiring shall be insulated from the instrument case.**

3.2.9 Equipment safety - Exposed circuitry should be avoided. When this is impractical, the instrument shall be constructed to provide maximum protection from accidental short circuits. Special attention shall be given to circuitry design to insure that electrical equipment is adequately protected from possible damage caused by movement of personnel, maintenance handling, and loose or stored materials. Lamp sockets and holders in instrument lighting panels shall be designed and constructed with exposed conducting surfaces at ground potential.

3.2.10 Interference control - The generation of radio interference by the instrument and the vulnerability of the instrument to radio interference shall be controlled within the limits of MIL-I-6181.

3.2.11 Installation instructions - The contractor shall furnish with each instrument one copy of the installation instructions with illustrations and diagrams as necessary. Prior to printing, two copies shall be furnished to the procuring activity for approval. Whenever possible the instructions shall be arranged to require only one sheet of paper either 8-1/2 by 11 or 11 by 17 inches.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:34Z
Check the source to verify that this is the currentversion before use.

3.2.12 Envelope - An envelope, furnished by the contractor containing materials as labeled, shall be packaged with each instrument and shall be marked.

IMPORTANT
THIS ENVELOPE CONTAINS
INSTALLATION INSTRUCTIONS

3.2.13 Power input - The instrument shall be designed to operate from power sources with characteristics conforming to MIL-STD-704 and the detail specification.

3.2.13.1 Undervoltage protection - The instrument shall not be damaged by voltages below the minimum specified and shall automatically resume normal operation when the voltage returns within the specified limits.

3.2.13.2 Overload protection - Overload protection shall not be accomplished by protective devices such as fuses, circuit breakers, etc., unless approved by the procuring activity. All parts and circuits which are likely to carry an overload due to failures or poor adjustments shall be proportioned to withstand such overload without damage to the instrument. The instrument shall resume normal operation when the voltage returns within the specified limits.

3.2.14 Airspeed instruments - The standard pressure used in calibrating indicators for use with pitot-static tubes shall be in accordance with the values specified in ANA-418.

3.2.14.1 Application - The pitot pressure shall be applied to the inside of the diaphragm capsule and the static pressure to the interior of the case.

3.2.14.2 Capillary tubing - A sufficient length of approved capillary tubing having a minimum inside diameter of .020 inch, or some other acceptable restriction, shall be incorporated in the instrument case between the pitot connection in the rear of the case and the diaphragm capsule. The effect of this restriction shall be such that the indicator will meet the requirements of the "damping tests" specified herein and in the detail specification.

3.2.14.3 Plug - Each static and pitot pressure connection shall be provided with a vented plug of material and a design acceptable to the procuring activity. The vent hole shall be 1/16 ± 1/64 inch in diameter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:34Z
Check the source to verify that this is the current version before use.

3.2.15 Warmup time - Unless otherwise specified by the detail specification, the warmup time at temperatures down to -54° C shall be not more than 2 minutes for equipment essential to flight safety, and shall be not more than 5 minutes for equipment not essential to flight safety.

3.2.16 Service conditions (environmental) - The instrument shall be so designed and constucted that no fixed part or assembly shall become loose, no moving or movable part or assembly become undesirably free or sluggish in operation, no movable part or control be shifted in setting, position, or adjustment, and no degradation be caused in the performance below that specified in the individual specification during operation or after storage in the environmental conditions, or any natural combinations of these conditions as specified in the detail specification, or if not specified therein as follows:

3.2.16.1 Temperature -

(a) Operating - Continuous operation over the range of -54° to +71° C (-65° to +160° F), the temperature remaining constant for long periods or varying at a rate as high as 1° C per second.

(b) Nonoperating - The instrument in a nonoperating condition shall withstand long periods of exposure to the extremes of -62° to +95° C (-80° to +203° F)°.

3.2.16.2 Altitude - Continuous operation over the range of sea level to 70,000 feet, the pressure remaining constant or varying at a rate as high as 0.5 inch of mercury per second.

3.2.16.3 Temperature-Altitude - The instrument shall be exposed to the temperature-altitude conditions shown in Figure 1.

3.2.16.4 Vibration - The instrument shall operate satisfactorily when subjected to vibration within the frequency range and amplitude as shown in Figure 2.

3.2.16.5 Humidity - The instrument shall withstand the effects of humidities up to 100 percent, including conditions wherein condensation takes place in and on the equipment. The instrument shall withstand the above conditions during continuous operation, intermittent operations, short-time operations, and exposure in a nonoperating condition.

3.2.16.6 Shock - The instrument shall not suffer damage or subsequently fail to provide the performance specified in the detail specification when subjected to 18 impact shocks of 15g, consisting of 3 shocks in opposite directions along each of 3 mutually perpendicular axes, each shock impulse having a time duration of 11 ± 1 milliseconds. The "g" value shall be within ± 10 percent when measured with a 0.2- to 250-cps filter, and the maximum "g" shall occur at approximately 5-1/2 milliseconds.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:34Z
Check the source to verify that this is the current version before use.

3.2.16.7 Salt atmosphere - The instrument shall withstand, in an operating and nonoperating condition, extended exposure to salt-sea atmosphere.

3.2.16.8 Fungus - The instrument shall withstand, in both an operating and nonoperating condition, extended exposure to fungus growth as encountered in tropical climates.

3.2.16.9 Sand and dust - The instrument shall withstand, in both an operating and nonoperating condition, extended exposure to sand and dust particles as encountered in desert areas.

3.3 Interchangeability -

3.3.1 Interchangeability shall be in accordance with MIL-STD-454 Requirement 7.

3.3.2 Interchangeability conflicts - Should any conflicts arise between any of the requirements of this specification or the detail specification and the requirement for interchangeability the contractor should immediately inform the procuring activity of such conflict. Unless advised otherwise, the interchangeability requirements shallgovern.

3.4

3.4.1 Nomenclature and nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with MIL-N-18307.

3.4.2 Nameplate - A nameplate conforming to MIL-N-18307 shall be securely attached to the exterior of the case and shall be durably marked to withstand usage encountered in service.

3.4.3 Manufacturer's part number - The manufacturer's part number on the nameplate shall be identical with the manufacturer's engineering production drawing number including applicable dash numbers if the drawing is tabulated and covers more than one part.

3.5 Reliability, longevity, and maintainability - The reliability, longevity, and maintainability requirements shall be as specified in the detail equipment specification. Where no military detail specification exists, the reliability, longevity, and maintainability requirements shall be negotiated with the procuring activity.

3.6 Workmanship - Workmanship shall be in accordance with MIL-STD-454, Requirement 9, Workmanship.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:34Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance provisions shall be as specified in each detail specification.

5. PREPARATION FOR DELIVERY

5.1 Packaging - The instrument shall be packaged in accordance with MIL-STD-794, Level A or C as specified in contract or order. For Level A packaging, the method of preservation shall be in accordance with MIL-P-16, Method 1A-5, without preservative compound, using metal reusable containers.

5.2 Packing - The instrument shall be packed in accordance with MIL-STD-794, Level A, B, or C as specified in contract or order.

5.3 Marking - The interior and exterior containers shall be marked as specified in MIL-STD-794 with the additional precautionary marking

FRAGILE
DELICATE INSTRUMENT
HANDLE WITH CARE

appearing conspicuously on two opposite sides of each interior package and exterior shipping container wherever practical.

6. NOTES

6.1 Intended use - This specification is intended for use to incorporate, by reference in detail specification, those requirements which are common to aircraft instruments.

6.2 Definitions -

6.2.1 Double amplitude - Whenever the words "double amplitude" are used, they shall mean the extent of motion as measured from one extreme to the opposite extreme.

6.2.2 Good as new - "Good as new" shall mean the aircraft instrument operated less than a percentage of specified MTBF operation, as specified in the detail specification.

6.2.3 Refurbished - "Refurbished" shall mean the aircraft instrument has been completely overhauled with all component parts meeting current parts standards, and the aircraft instrument shall have been subjected to, and met, all the requirements of a new aircraft instrument.

6.2.4 Military designations - Military designations shall not be applied to a product, except for Qualification/Production Test Samples, nor referred to in correspondence or sales matter, until notification has been received from the activity responsible for qualification/production testing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:34Z
Check the source to verify that this is the current version before use.

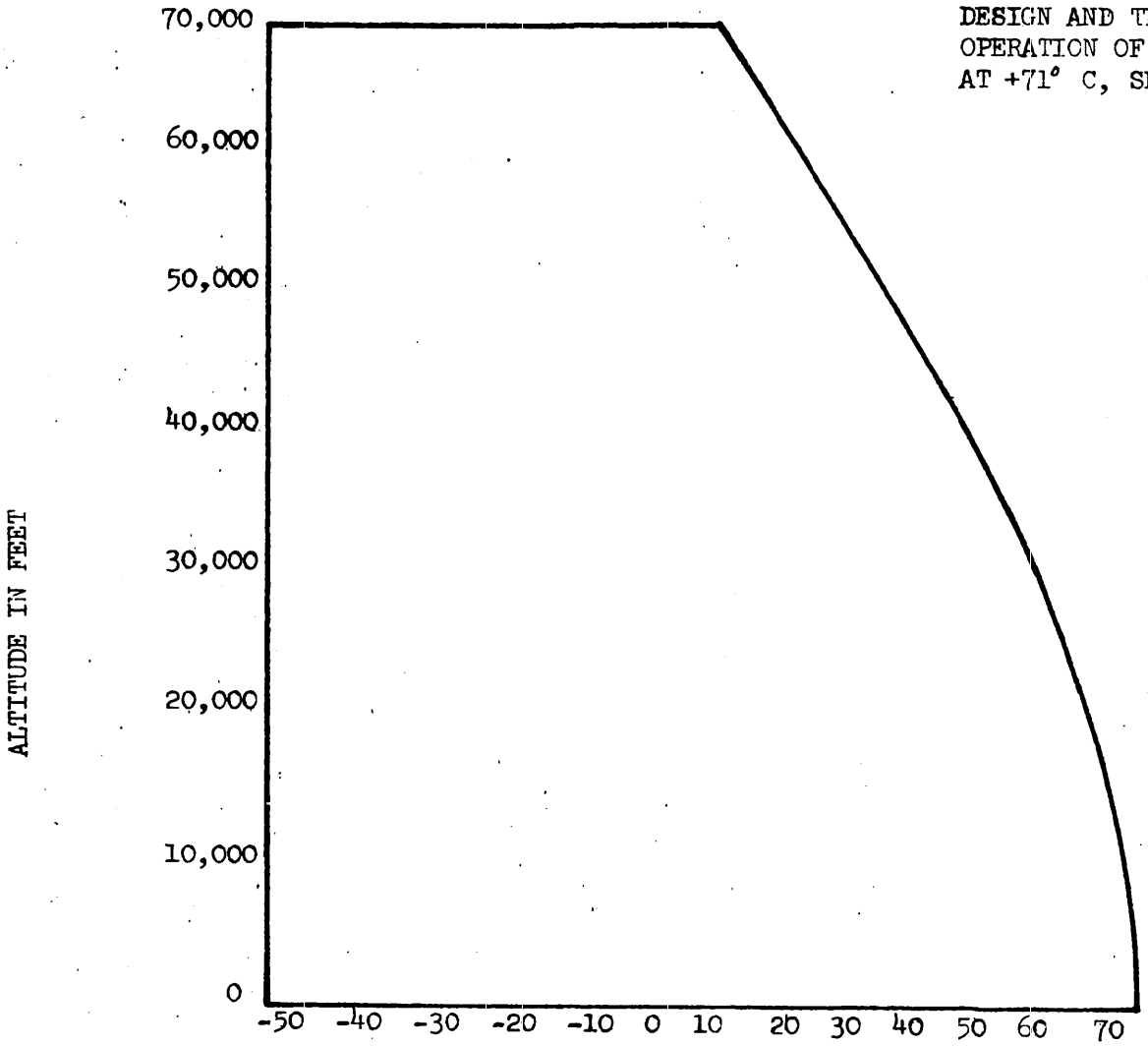


Figure 1. Operational requirements for Class 2 airborne electronic equipment - Temperature vs Altitude

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:34Z
Check the source to verify that this is the current version before use.

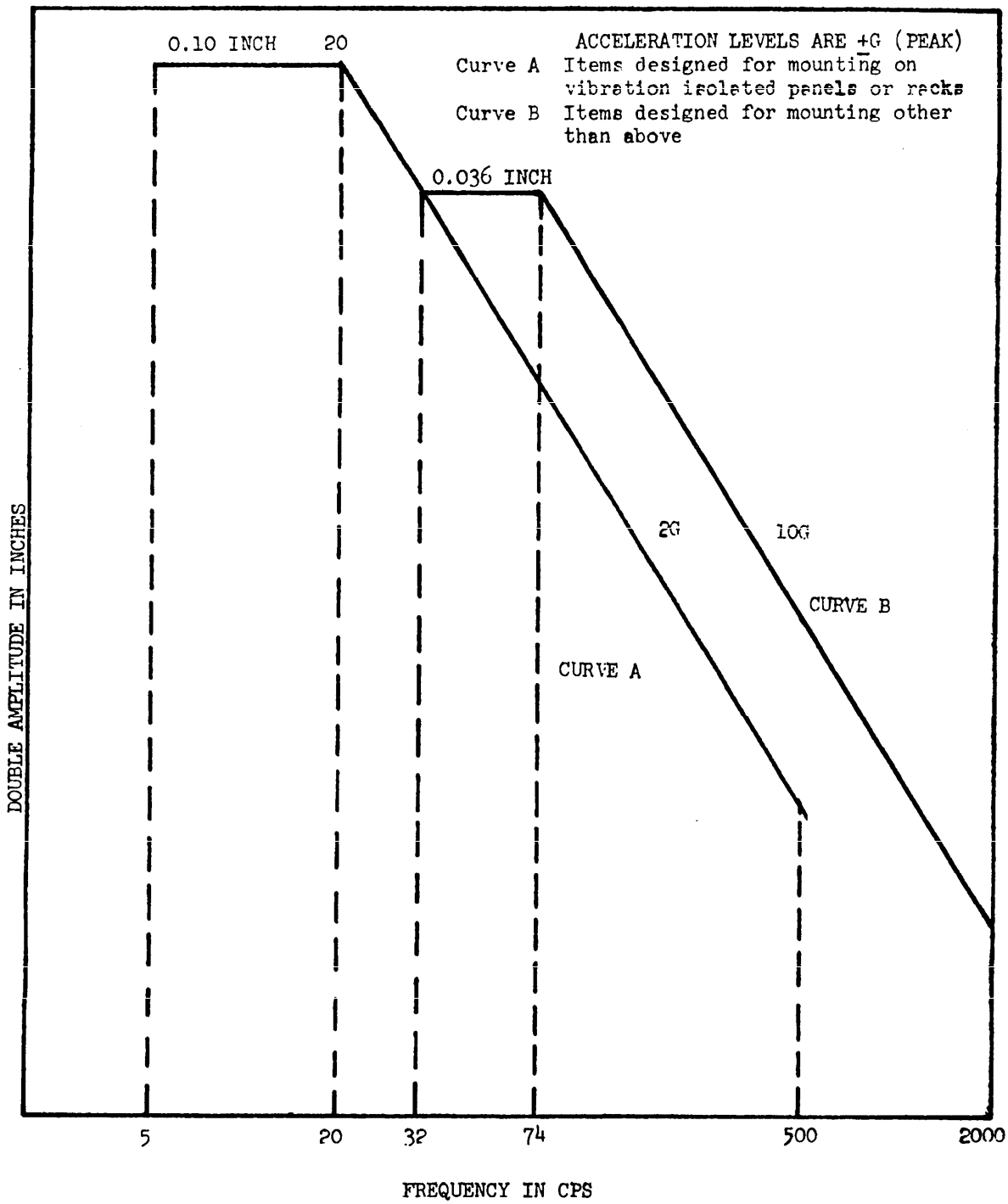


Figure 2. Vibration test curves, aircraft instruments

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:34Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:34Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-I-81400(AS) INSTRUMENTS AIRCRAFT GENERAL SPEC FOR

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD, FORM 1 APR 63 1426

C-8378

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:34Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID

U.S.MAIL

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

Commanding Officer
Naval Air Engineering Center
Engineering Specifications and Standards
Department (ESSD) Code 93
Lakehurst, NJ 08733

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:34Z
Check the source to verify that this is the current version before use.